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METHOD FOR TRANSMITTING GRAPHICAL USER INTERFACE DATA AND CONTROL SIGNALS TO A DISPARATE DEVICE

Abstract

A method for providing recommendations to a user for reducing spending in response to the user identifying how much to save and considering user spending preferences. The method includes allowing the user to enter how much the user wants to save; allowing the user to enter information concerning areas that the user does not want to reduce spending; determining what the user is spending on, where the user is spending, how much the user is spending and how often the user is spending; providing recommendations to the user of areas that the user can reduce spending; and prioritizing the recommendations for reducing spending by first recommending areas for reducing spending on things that the user has not identified as being areas that the user does not want to reduce spending and then recommending areas for reducing spending on areas that the user does not want to reduce spending.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. application Ser. No. 18/163,911, titled, Computer Network Back-End System for Transmitting Graphical User Interface Data and Control Signals to a Disparate Device, filed Feb. 2, 2023, whose content is hereby incorporated by reference.

BACKGROUND

Field

[0002] This disclosure relates generally to a method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save and, more particularly, to a method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save and considering user spending preferences.

Discussion

[0003] A bank is a financial institution that is licensed to receive deposits from individuals and organizations and to make loans to those individuals and organizations or others. Banks may also perform other services such as wealth management, currency exchange, etc. Therefore, a bank may have thousands of customers and clients. Depending on the services that a bank provides, it may be classified as a retail bank, a commercial bank, an investment bank or some combination thereof. A retail bank typically provides services such as checking and savings accounts, loan and mortgage services, financing for automobiles, and short-term loans such as overdraft protection. A commercial bank typically provides credit services, cash management, commercial real estate services, employer services, trade finance, etc. An investment bank typically provides corporate clients with complex services and financial transactions such as underwriting and assisting with merger and acquisition activity.

[0004] Most, and maybe all, banks provide systems, software and applications for online or mobile banking that allows customers and users of the bank to access their accounts through the internet on, for example, a smart phone, tablet or computer to perform certain tasks, such as seeing account balances and perform transactions, such as bill paying, funds transfer, check deposit, etc., without having to visit the bank or call the bank. Depending on the particular bank, those applications may provide many features and functions or may be limited to simple transactions.

SUMMARY

[0005] The following discussion discloses and describes a method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save and considering user spending preferences. The method includes allowing the user to enter how much the user wants to save; allowing the user to enter information concerning areas that the user does not want to reduce spending; determining what the user is spending on, where the user is spending, how much the user is spending and how often the user is spending; providing recommendations to the user of areas that the user can reduce spending considering the amount the user wants to save and the areas that the user does not want to reduce spending; and prioritizing the recommendations for reducing spending by first recommending areas for reducing spending on things that the user has not identified as being areas that the user does not want to reduce spending

and then recommending areas for reducing spending on areas that the user does not want to reduce spending.

[0006] Additional features of the disclosure will become apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 illustrates a system and environment thereof by which a user benefits through use of services and products of an enterprise system;

[0008] FIG. 2 is a diagram of a feedforward network;

[0009] FIG. 3 is a diagram of a convolutional neural network (CNN);

[0010] FIG. 4 is a diagram of a portion of the CNN shown in FIG. 3 illustrating assigned weights at connections or neurons;

[0011] FIG. 5 is a diagram representing an exemplary weighted sum computation in a node in an artificial neural network;

[0012] FIG. 6 is a diagram of a recurrent neural network (RNN) utilized in machine learning;

[0013] FIG. 7 is a schematic logic diagram of an artificial intelligence processor operating an artificial intelligence program;

[0014] FIG. 8 is a flow chart showing a method for model development and deployment by machine learning;

[0015] FIG. 9 is a simple illustration of a typical landing page for an online or mobile banking application;

[0016] FIG. 10 is a simple illustration of a page in the application that may open in response to a user selecting a save money button;

[0017] FIG. 11 is a simple illustration of a page showing a user conversing with a bot; and

[0018] FIG. 12 is a block diagram of an architecture that could be part of a system that operates the online or mobile banking application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0019] The following discussion of the embodiments of the disclosure directed to a method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save and considering user spending preferences is merely exemplary in nature, and is in no way intended to limit the disclosure or its applications or uses.

[0020] Embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the disclosure are shown. Indeed, the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout. Unless described or implied as exclusive alternatives, features throughout the drawings and descriptions should be taken as cumulative, such that features expressly associated with some particular embodiments can be combined with other embodiments. Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which the presently disclosed subject matter pertains.

[0021] The exemplary embodiments are provided so that this disclosure will be both thorough and complete, and will fully convey the scope of the disclosure and enable one of ordinary skill in the art to make, use and practice the disclosure.

[0022] The terms “coupled,” “fixed,” “attached to,” “communicatively coupled to,” “operatively coupled to,” and the like refer to both (i) direct connecting, coupling, fixing, attaching, communicatively coupling; and (ii) indirect connecting coupling, fixing, attaching,

communicatively coupling via one or more intermediate components or features, unless otherwise specified herein. “Communicatively coupled to” and “operatively coupled to” can refer to physically and/or electrically related components.

[0023] Embodiments of the present disclosure described herein, with reference to flowchart illustrations and/or block diagrams of methods or apparatuses (the term “apparatus” includes systems and computer program products), will be understood such that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a particular machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create mechanisms for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0024] These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer readable memory produce an article of manufacture including instructions, which implement the function/act specified in the flowchart and/or block diagram block or blocks.

[0025] The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions, which execute on the computer or other programmable apparatus, provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. Alternatively, computer program implemented steps or acts may be combined with operator or human implemented steps or acts in order to carry out an embodiment of the disclosure.

[0026] While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad disclosure, and that this disclosure not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations, modifications, and combinations of the herein described embodiments can be configured without departing from the scope and spirit of the disclosure. Therefore, it is to be understood that, within the scope of the included claims, the disclosure may be practiced other than as specifically described herein.

[0027] FIG. 1 illustrates a system **10**, such as a banking system, and environment thereof by which a user **18** benefits through use of services and products of an enterprise system **12**. The environment may include, for example, a distributed cloud computing environment, such as private cloud, public cloud, community cloud and/or hybrid cloud, and on-premise environment, fog computing environment and/or an edge computing environment. The user **18** accesses services and products by use of one or more user devices, illustrated in separate examples as a computing device **14** and a mobile device **16**, which may be, as non-limiting examples, a smart phone, a portable digital assistant (PDA), a pager, a mobile television, a gaming device, a laptop computer, a camera, a video recorder, an audio/video player, radio, a GPS device, or any combination of the aforementioned, or other portable device with processing and communication capabilities. In the illustrated example, the mobile device **16** is the system **10** as having exemplary elements, the below descriptions of which apply as well to the computing device **14**, which can be, as non-limiting examples, a desktop computer, a laptop computer or other user-accessible computing device.

[0028] Furthermore, the user device, referring to either or both of the computing device **14** and the mobile device **16**, may be or include a workstation, a server, or any other suitable device, including a set of servers, a cloud-based application or system, or any other suitable system, adapted to

execute, for example any suitable operating system including Linux, UNIX, Windows, macOS, iOS, Android and any other known operating system used on personal computers, central computing systems, phones, and other devices.

[0029] The user **18** can be an individual, a group, or any entity in possession of or having access to the user device, referring to either or both of the computing device **14** and the mobile device **16**, which may be personal or public items. Although the user **18** may be singly represented in some drawings, at least in some embodiments according to these descriptions the user **18** is one of many such that a market or community of users, consumers, customers, business entities, government entities, clubs, and groups of any size are all within the scope of these descriptions.

[0030] The user device, as illustrated with reference to the mobile device **16**, includes components such as at least one of each of a processing device **20**, and a memory device **22** for processing use, such as random access memory (RAM), and read-only memory (ROM). The illustrated mobile device **16** further includes a storage device **24** including at least one of a non-transitory storage medium, such as a microdrive, for long-term, intermediate-term, and short-term storage of computer-readable instructions **26** for execution by the processing device **20**. For example, the instructions **26** can include instructions for an operating system and various applications or programs **30**, of which the application **32** is represented as a particular example. The storage device **24** can store various other data items **34**, which can include, as non-limiting examples, cached data, user files such as those for pictures, audio and/or video recordings, files downloaded or received from other devices, and other data items preferred by the user or required or related to any or all of the applications or programs **30**.

[0031] The memory device **22** is operatively coupled to the processing device **20**. As used herein, memory includes any computer readable medium to store data, code, or other information. The memory device **22** may include volatile memory, such as volatile RAM including a cache area for the temporary storage of data. The memory device **22** may also include non-volatile memory, which can be embedded and/or may be removable. The non-volatile memory can additionally or alternatively include an electrically erasable programmable read-only memory (EEPROM), flash memory or the like.

[0032] According to various embodiments, the memory device **22** and the storage device **24** may be combined unto a single medium. The memory device **22** and the storage device **24** can store any of a number of applications that comprise computer-executable instructions and code executed by the processing device **20** to implement the functions of the mobile device **16** described herein. For example, the memory device **22** may include such applications as a conventional web browser application and/or a mobile P2P payment system client application. These applications also typically provide a graphical user interface (GUI) on a display **40** that allows the user **18** to communicate with the mobile device **16**, and, for example, a mobile banking system, and/or other devices or systems. In one embodiment, when the user **18** decides to enroll in a mobile banking program, the user **18** downloads or otherwise obtains the mobile banking system client application from a mobile banking system, for example, the enterprise system **12**, or from a distinct application server. In other embodiments, the user **18** interacts with a mobile banking system via a web browser application in addition to, or instead of, the mobile P2P payment system client application.

[0033] The processing device **20**, and other processors described herein, generally include circuitry for implementing communication and/or logic functions of the mobile device **16**. For example, the processing device **20** may include a digital signal processor, a microprocessor, and various analog to digital converters, digital to analog converters, and/or other support circuits. Control and signal processing functions of the mobile device **16** are allocated between these devices according to their respective capabilities. The processing device **20** thus may also include the functionality to encode and interleave messages and data prior to modulation and transmission. The processing device **20** can additionally include an internal data modem. Further, the processing device **20** may include functionality to operate one or more software programs, which may be stored in the memory device

22, or in the storage device **24**. For example, the processing device **20** may be capable of operating a connectivity program, such as a web browser application. The web browser application may then allow the mobile device **16** to transmit and receive web content, such as, for example, location-based content and/or other web page content, according to a wireless application protocol (WAP), hypertext transfer protocol (HTTP), and/or the like.

[0034] The memory device **22** and the storage device **24** can each also store any of a number of pieces of information, and data, used by the user device and the applications and devices that facilitate functions of the user device, or are in communication with the user device, to implement the functions described herein and others not expressly described. For example, the storage device **24** may include such data as user authentication information, etc.

[0035] The processing device **20**, in various examples, can operatively perform calculations, can process instructions for execution and can manipulate information. The processing device **20** can execute machine-executable instructions stored in the storage device **24** and/or the memory device **22** to thereby perform methods and functions as described or implied herein, for example, by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subject matters of these descriptions pertain. The processing device **20** can be or can include, as non-limiting examples, a central processing unit (CPU), a microprocessor, a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a programmable logic device (PLD), a digital signal processor (DSP), a field programmable gate array (FPGA), a state machine, a controller, gated or transistor logic, discrete physical hardware components, and combinations thereof. In some embodiments, particular portions or steps of methods and functions described herein are performed in whole or in part by way of the processing device **20**, while in other embodiments methods and functions described herein include cloud-based computing in whole or in part such that the processing device **20** facilitates local operations including, as non-limiting examples, communication, data transfer, and user inputs and outputs such as receiving commands from and providing displays to the user.

[0036] The mobile device **16**, as illustrated, includes an input and output system **36**, referring to, including, or operatively coupled with, one or more user input devices and/or one or more user output devices, which are operatively coupled to the processing device **20**. The input and output system **36** may include input/output circuitry that may operatively convert analog signals and other signals into digital data, or may convert digital data to another type of signal. For example, the input/output circuitry may receive and convert physical contact inputs, physical movements or auditory signals (e.g., which may be used to authenticate a user) to digital data. Once converted, the digital data may be provided to the processing device **20**. The input and output system **36** may also include the display **40** (e.g., a liquid crystal display (LCD), light emitting diode (LED) display or the like), which can be, as a non-limiting example, a presence-sensitive input screen (e.g. touch screen or the like) of the mobile device **16**, which serves both as an output device, by providing graphical and text indicia and presentations for viewing by one or more of the users **18**, and as an input device, by providing virtual buttons, selectable options, a virtual keyboard, and other indicia that, when touched, control the mobile device **16** by user action. The user output devices include a speaker **44** or other audio device. The user input devices, which allow the mobile device **16** to receive data and actions such as button manipulations and touches from a user such as the user **18**, may include any of a number of devices allowing the mobile device **16** to receive data from a user, such as a keypad, keyboard, touch-screen, touchpad, microphone **42**, mouse, joystick, other pointer device, button, soft key, infrared sensor and/or other input device(s). The input and output system **36** may also include a camera **46**, such as a digital camera.

[0037] Further non-limiting examples of input devices and output devices include one or more of each, any and all of a wireless or wired keyboard, a mouse, a touchpad, a button, a switch, a light, an LED, a buzzer, a bell, a printer and/or other user input devices and output devices for use by or communication with the user **18** in accessing, using, and controlling, in whole or in part, the user

device, referring to either or both of the computing device **14** and the mobile device **16**. Inputs by one or more of the users **18** can thus be made via voice, text or graphical indicia selections. For example, such inputs in some examples correspond to user-side actions and communications seeking services and products of the enterprise system **12**, and at least some outputs in such examples correspond to data representing enterprise-side actions and communications in two-way communications between the user **18** and the enterprise system **12**.

[0038] The input and output system **36** may also be configured to obtain and process various forms of authentication via an authentication system to obtain authentication information of the user **18**. Various authentication systems may include, according to various embodiments, a recognition system that detects biometric features or attributes of the user **18**, such as fingerprint recognition systems, handprint recognition systems, palm print recognition systems, iris recognition systems, facial recognition systems, speech recognition systems, DNA-based authentication or any other suitable biometric attribute or information associated with the user **18**. Alternate authentication systems may include one or more systems to identify the user **18** based on a visual or temporal pattern of inputs provided by the user **18**. For example, the user device may display selectable options, shapes, inputs, buttons, numeric representations, etc. that may be selected in a predetermined specified order or according to a specific pattern. Other authentication processes are also contemplated herein, including email authentication, password protected authentication, phone call authentication, etc. The user device may enable the user **18** to input any number or combination of authentication systems.

[0039] The user device, referring to either or both of the computing device **14** or the mobile device **16** may also include a positioning device **48**, which can be, for example, a global positioning system (GPS) device configured to be used by a positioning system to determine a location of the computing device **14** or the mobile device **16**. For example, the positioning system device **48** may include a GPS transceiver. In some embodiments, the positioning system device **48** includes an antenna, transmitter, and receiver. For example, in one embodiment, triangulation of cellular signals may be used to identify the approximate location of the mobile device **16**. In other embodiments, the positioning device **48** includes a proximity sensor or transmitter, such as an RFID tag, that can sense or be sensed by devices known to be located proximate a merchant or other location to determine that the consumer mobile device **16** is located proximate these known devices.

[0040] In the illustrated example, a system intraconnect **38**, connects, for example electrically, the various described, illustrated, and implied components of the mobile device **16**. The intraconnect **38**, in various non-limiting examples, can include or represent, a system bus, a high-speed interface connecting the processing device **20** to the memory device **22**, individual electrical connections among the components, and electrical conductive traces on a motherboard common to some or all of the above-described components of the user device. As discussed herein, the system intraconnect **38** may operatively couple various components with one another, or in other words, electrically connects those components, either directly or indirectly—by way of intermediate component(s)—with one another.

[0041] The user device, referring to either or both of the computing device **14** and the mobile device **16**, with particular reference to the mobile device **16** for illustration purposes, includes a communication interface **50**, by which the mobile device **16** communicates and conducts transactions with other devices and systems. The communication interface **50** may include digital signal processing circuitry and may provide two-way communications and data exchanges, for example, wirelessly via wireless communication device **52**, and for an additional or alternative example, via wired or docked communication by mechanical electrically conductive connector **54**. Communications may be conducted via various modes or protocols, of which GSM voice calls, SMS, EMS, MMS messaging, TDMA, CDMA, PDC, WCDMA, CDMA2000, and GPRS, are all non-limiting and non-exclusive examples. Thus, communications can be conducted, for example,

via the wireless communication device **52**, which can be or include a radio-frequency transceiver, a Bluetooth device, Wi-Fi device, a near-field communication device, and other transceivers. In addition, GPS may be included for navigation and location-related data exchanges, ingoing and/or outgoing. Communications may also or alternatively be conducted via the connector **54** for wired connections such by USB, Ethernet, and other physically connected modes of data transfer.

[0042] The processing device **20** is configured to use the communication interface **50** as, for example, a network interface to communicate with one or more other devices on a network. In this regard, the communication interface **50** utilizes the wireless communication device **52** as an antenna operatively coupled to a transmitter and a receiver (together a “transceiver”) included with the communication interface **50**. The processing device **20** is configured to provide signals to and receive signals from the transmitter and receiver, respectively. The signals may include signaling information in accordance with the air interface standard of the applicable cellular system of a wireless telephone network. In this regard, the mobile device **16** may be configured to operate with one or more air interface standards, communication protocols, modulation types, and access types. By way of illustration, the mobile device **16** may be configured to operate in accordance with any of a number of first, second, third, fourth or fifth-generation communication protocols and/or the like. For example, the mobile device **16** may be configured to operate in accordance with second-generation (2G) wireless communication protocols IS-136 (time division multiple access (TDMA)), GSM (global system for mobile communication), and/or IS-95 (code division multiple access (CDMA)), or with third-generation (3G) wireless communication protocols, such as universal mobile telecommunications System (UMTS), CDMA2000, wideband CDMA (WCDMA) and/or time division-synchronous CDMA (TD-SCDMA), with fourth-generation (4G) wireless communication protocols such as long-term evolution (LTE), fifth-generation (5G) wireless communication protocols, Bluetooth low energy (BLE) communication protocols such as Bluetooth 5.0, ultra-wideband (UWB) communication protocols, and/or the like. The mobile device **16** may also be configured to operate in accordance with non-cellular communication mechanisms, such as via a wireless local area network (WLAN) or other communication/data networks.

[0043] The communication interface **50** may also include a payment network interface. The payment network interface may include software, such as encryption software, and hardware, such as a modem, for communicating information to and/or from one or more devices on a network. For example, the mobile device **16** may be configured so that it can be used as a credit or debit card by, for example, wirelessly communicating account numbers or other authentication information to a terminal of the network. Such communication could be performed via transmission over a wireless communication protocol such as the Near-field communication protocol.

[0044] The mobile device **16** further includes a power source **28**, such as a battery, for powering various circuits and other devices that are used to operate the mobile device **16**. Embodiments of the mobile device **16** may also include a clock or other timer configured to determine and, in some cases, communicate actual or relative time to the processing device **20** or one or more other devices. For a further example, the clock may facilitate timestamping transmissions, receptions, and other data for security, authentication, logging, polling, data expiry and forensic purposes.

[0045] The system **10** as illustrated diagrammatically represents at least one example of a possible implementation, where alternatives, additions, and modifications are possible for performing some or all of the described methods, operations and functions. Although shown separately, in some embodiments, two or more systems, servers, or illustrated components may be utilized. In some implementations, the functions of one or more systems, servers, or illustrated components may be provided by a single system or server. In some embodiments, the functions of one illustrated system or server may be provided by multiple systems, servers, or computing devices, including those physically located at a central facility, those logically local, and those located as remote with respect to each other.

[0046] The enterprise system **12** can offer any number or type of services and products to one or

more of the users **18**. In some examples, the enterprise system **12** offers products, and in some examples, the enterprise system **12** offers services. Use of “service(s)” or “product(s)” thus relates to either or both in these descriptions. With regard, for example, to online information and financial services, “service” and “product” are sometimes termed interchangeably. In non-limiting examples, services and products include retail services and products, information services and products, custom services and products, predefined or pre-offered services and products, consulting services and products, advising services and products, forecasting services and products, internet products and services, social media, and financial services and products, which may include, in non-limiting examples, services and products relating to banking, checking, savings, investments, credit cards, automatic-teller machines, debit cards, loans, mortgages, personal accounts, business accounts, account management, credit reporting, credit requests and credit scores.

[0047] To provide access to, or information regarding, some or all the services and products of the enterprise system **12**, automated assistance may be provided by the enterprise system **12**. For example, automated access to user accounts and replies to inquiries may be provided by enterprise-side automated voice, text, and graphical display communications and interactions. In at least some examples, any number of human agents **60**, can be employed, utilized, authorized or referred by the enterprise system **12**. Such human agents **60** can be, as non-limiting examples, point of sale or point of service (POS) representatives, online customer service assistants available to the users **18**, advisors, managers, sales team members, and referral agents ready to route user requests and communications to preferred or particular other agents, human or virtual.

[0048] The human agents **60** may utilize agent devices **62** to serve users in their interactions to communicate and take action. The agent devices **62** can be, as non-limiting examples, computing devices, kiosks, terminals, smart devices such as phones, and devices and tools at customer service counters and windows at POS locations. In at least one example, the diagrammatic representation of the components of the mobile device **16** in FIG. **1** applies as well to one or both of the computing device **14** and the agent devices **62**.

[0049] The agent devices **62** individually or collectively include input devices and output devices, including, as non-limiting examples, a touch screen, which serves both as an output device by providing graphical and text indicia and presentations for viewing by one or more of the agents **60**, and as an input device by providing virtual buttons, selectable options, a virtual keyboard, and other indicia that, when touched or activated, control or prompt the agent device **62** by action of the attendant agent **60**. Further non-limiting examples include, one or more of each, any, and all of a keyboard, a mouse, a touchpad, a joystick, a button, a switch, a light, an LED, a microphone serving as input device for example for voice input by the human agent **60**, a speaker serving as an output device, a camera serving as an input device, a buzzer, a bell, a printer and/or other user input devices and output devices for use by or communication with the human agent **60** in accessing, using, and controlling, in whole or in part, the agent device **62**.

[0050] Inputs by one or more of the human agents **60** can thus be made via voice, text or graphical indicia selections. For example, some inputs received by the agent device **62** in some examples correspond to, control, or prompt enterprise-side actions and communications offering services and products of the enterprise system **12**, information thereof, or access thereto. At least some outputs by the agent device **62** in some examples correspond to, or are prompted by, user-side actions and communications in two-way communications between the user **18** and an enterprise-side human agent **60**.

[0051] From a user perspective experience, an interaction in some examples within the scope of these descriptions begins with direct or first access to one or more of the human agents **60** in person, by phone or online for example via a chat session or website function or feature. In other examples, a user is first assisted by a virtual agent **64** of the enterprise system **12**, which may satisfy user requests or prompts by voice, text or online functions, and may refer users to one or more of the human agents **60** once preliminary determinations or conditions are made or met.

[0052] The enterprise system **12** includes a computing system **70** having various components, such as a processing device **72** and a memory device **74** for processing use, such as random access memory (RAM) and read-only memory (ROM). The computing system **70** further includes a storage device **76** having at least one non-transitory storage medium, such as a microdrive, for long-term, intermediate-term, and short-term storage of computer-readable instructions **78** for execution by the processing device **72**. For example, the instructions **78** can include instructions for an operating system and various applications or programs **80**, of which an application **82** is represented as a particular example. The storage device **76** can store various other data **84**, which can include, as non-limiting examples, cached data, and files such as those for user accounts, user profiles, account balances, and transaction histories, files downloaded or received from other devices, and other data items preferred by the user or required or related to any or all of the applications or programs **80**.

[0053] The computing system **70**, in the illustrated example, also includes an input/output system **86**, referring to, including, or operatively coupled with input devices and output devices such as, in a non-limiting example, agent devices **62**, which have both input and output capabilities.

[0054] In the illustrated example, a system intraconnect **88** electrically connects the various above-described components of the computing system **70**. In some cases, the intraconnect **88** operatively couples components to one another, which indicates that the components may be directly or indirectly connected, such as by way of one or more intermediate components. The intraconnect **88**, in various non-limiting examples, can include or represent, a system bus, a high-speed interface connecting the processing device **72** to the memory device **74**, individual electrical connections among the components, and electrical conductive traces on a motherboard common to some or all of the above-described components of the user device.

[0055] The computing system **70** includes a communication interface **90** by which the computing system **70** communicates and conducts transactions with other devices and systems. The communication interface **90** may include digital signal processing circuitry and may provide two-way communications and data exchanges, for example wirelessly via wireless device **92**, and for an additional or alternative example, via wired or docked communication by mechanical electrically conductive connector **94**. Communications may be conducted via various modes or protocols, of which GSM voice calls, SMS, EMS, MMS messaging, TDMA, CDMA, PDC, WCDMA, CDMA2000, and GPRS, are all non-limiting and non-exclusive examples. Thus, communications can be conducted, for example, via the wireless device **92**, which can be or include a radio-frequency transceiver, a Bluetooth device, Wi-Fi device, near-field communication device, and other transceivers. In addition, GPS may be included for navigation and location-related data exchanges, ingoing and/or outgoing. Communications may also or alternatively be conducted via the connector **94** for wired connections such as by USB, Ethernet, and other physically connected modes of data transfer.

[0056] The processing device **72**, in various examples, can operatively perform calculations, can process instructions for execution, and can manipulate information. The processing device **72** can execute machine-executable instructions stored in the storage device **76** and/or the memory device **74** to thereby perform methods and functions as described or implied herein, for example by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subjects matters of these descriptions pertain. The processing device **72** can be or can include, as non-limiting examples, a central processing unit (CPU), a microprocessor, a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a programmable logic device (PLD), a digital signal processor (DSP), a field programmable gate array (FPGA), a state machine, a controller, gated or transistor logic, discrete physical hardware components, and combinations thereof.

[0057] Furthermore, the computing system **70**, may be or include a workstation, a server, or any other suitable device, including a set of servers, a cloud-based application or system, or any other

suitable system, adapted to execute, for example any suitable operating system, including Linux, UNIX, Windows, macOS, iOS, Android, and any known other operating system used on personal computer, central computing systems, phones and other devices.

[0058] The user devices, referring to either or both of the mobile device **16** and the computing device **14**, the agent devices **62** and the computing system **70**, which may be one or any number centrally located or distributed, are in communication through one or more networks, referenced as the system **10** in FIG. **1**.

[0059] The network **100** provides wireless or wired communications among the components of the network **100** and the environment thereof, including other devices local or remote to those illustrated, such as additional mobile devices, servers, and other devices communicatively coupled to the network **100**, including those not illustrated in FIG. **1**. The network **100** is singly depicted for illustrative convenience, but may include more than one network without departing from the scope of these descriptions. In some embodiments, the network **100** may be or provide one or more cloud-based services or operations. The network **100** may be or include an enterprise or secured network, or may be implemented, at least in part, through one or more connections to the Internet. A portion of the network **100** may be a virtual private network (VPN) or an Intranet. The network **100** can include wired and wireless links, including, as non-limiting examples, 802.11a/b/g/n/ac, 802.20, WiMax, LTE, and/or any other wireless link. The network **100** may include any internal or external network, networks, sub-network, and combinations of such operable to implement communications between various computing components within and beyond the network **100**. The network **100** may communicate, for example, internet protocol (IP) packets, frame relay frames, asynchronous transfer mode (ATM) cells, voice, video, data, and other suitable information between network addresses. The network **100** may also include one or more local area networks (LANs), radio access networks (RANs), metropolitan area networks (MANs), wide area networks (WANs), all or a portion of the internet and/or any other communication system or systems at one or more locations.

[0060] The network **100** may incorporate a cloud platform/data center that supports various service models including Platform-as-a-Service (PaaS), Infrastructure-as-a-Service (IaaS) and Software-as-a-Service (SaaS). Such service models may provide, for example, a digital platform accessible to the user device. Specifically, SaaS may provide the user **18** with the capacity to use applications running on a cloud infrastructure, where the applications are accessible via a thin client interface, such as a web browser, and the user **18** is not permitted to manage or control the underlying cloud infrastructure, i.e., network, servers, operating systems, storage or specific application capabilities that are not user specific. PaaS also does not permit the user **18** to manage or control the underlying cloud infrastructure, but this service may enable the user **18** to deploy user-created or acquired applications onto the cloud infrastructure using programming languages and tools provided by the provider of the application. In contrast, IaaS provides the user **18** the permission to provision processing, storage, networks and other computing resources as well as run arbitrary software such as operating systems and applications, thereby giving the user **18** control over operating systems, storage and deployed applications, and potentially select networking components, such as host firewalls.

[0061] The network **100** may also incorporate various cloud-based deployment models including private cloud, i.e., an organization-based cloud managed by either the organization or third parties and hosted on-premises or off premises, public cloud, i.e., cloud-based infrastructure available to the general public that is owned by an organization that sells cloud services, community cloud, i.e., cloud-based infrastructure shared by several organizations and managed by the organizations or third parties and hosted on-premises or off premises, and/or hybrid cloud, i.e., composed of two or more clouds e.g., private community and/or public.

[0062] Two external systems **102** and **104** are illustrated in FIG. **1** and representing any number and variety of data sources, users, consumers, customers, business entities, banking systems,

government entities, clubs, and groups of any size are all within the scope of the descriptions. In at least one example, the external systems **102** and **104** represent automatic teller machines (ATMs) utilized by the enterprise system **12** in serving the users **18**. In another example, the external systems **102** and **104** represent payment clearinghouse or payment rail systems for processing payment transactions, and in another example, the external systems **102** and **104** represent third party systems such as merchant systems configured to interact with the user device **16** during transactions and also configured to interact with the enterprise system **12** in back-end transactions clearing processes.

[0063] In certain embodiments, one or more of the systems such as the user device **16**, the enterprise system **12**, and/or the external systems **102** and **104** are, include, or utilize virtual resources. In some cases, such virtual resources are considered cloud resources or virtual machines. The cloud computing configuration may provide an infrastructure that includes a network of interconnected nodes and provides stateless, low coupling, modularity, and semantic interoperability. Such interconnected nodes may incorporate a computer system that includes one or more processors, a memory, and a bus that couples various system components (e.g., the memory) to the processor. Such virtual resources may be available for shared use among multiple distinct resource consumers and in certain implementations, virtual resources do not necessarily correspond to one or more specific pieces of hardware, but rather to a collection of pieces of hardware operatively coupled within a cloud computing configuration so that the resources may be shared as needed.

[0064] As used herein, an artificial intelligence system, artificial intelligence algorithm, artificial intelligence module, program, and the like, generally refer to computer implemented programs that are suitable to simulate intelligent behavior (i.e., intelligent human behavior) and/or computer systems and associated programs suitable to perform tasks that typically require a human to perform, such as tasks requiring visual perception, speech recognition, decision-making, translation, and the like. An artificial intelligence system may include, for example, at least one of a series of associated if-then logic statements, a statistical model suitable to map raw sensory data into symbolic categories and the like, or a machine learning program. A machine learning program, machine learning algorithm, or machine learning module, as used herein, is generally a type of artificial intelligence including one or more algorithms that can learn and/or adjust parameters based on input data provided to the algorithm. In some instances, machine learning programs, algorithms and modules are used at least in part in implementing artificial intelligence (AI) functions, systems and methods.

[0065] Artificial Intelligence and/or machine learning programs may be associated with or conducted by one or more processors, memory devices, and/or storage devices of a computing system or device. It should be appreciated that the artificial intelligence algorithm or program may be incorporated within the existing system architecture or be configured as a standalone modular component, controller, or the like communicatively coupled to the system. An artificial intelligence program and/or machine learning program may generally be configured to perform methods and functions as described or implied herein, for example by one or more corresponding flow charts expressly provided or implied as would be understood by one of ordinary skill in the art to which the subjects matters of these descriptions pertain.

[0066] A machine learning program may be configured to use various analytical tools such as algorithmic applications to leverage data to make predictions or decisions. Machine learning programs may be configured to implement various algorithmic processes and learning approaches including, for example, decision tree learning, association rule learning, artificial neural networks, recurrent artificial neural networks, long short term memory networks, inductive logic programming, support vector machines, clustering, Bayesian networks, reinforcement learning, representation learning, similarity and metric learning, sparse dictionary learning, genetic algorithms, k-nearest neighbor (KNN), and the like. In some embodiments, the machine learning

algorithm may include one or more image recognition algorithms suitable to determine one or more categories to which an input, such as data communicated from a visual sensor or a file in JPEG, PNG or other format, representing an image or portion thereof, belongs. Additionally or alternatively, the machine learning algorithm may include one or more regression algorithms configured to output a numerical value given an input. Further, the machine learning may include one or more pattern recognition algorithms, e.g., a module, subroutine or the like capable of translating text or string characters and/or a speech recognition module or subroutine. In various embodiments, the machine learning module may include a machine learning acceleration logic, e.g., a fixed function matrix multiplication logic, in order to implement the stored processes and/or optimize the machine learning logic training and interface.

[0067] Machine learning models are trained using various data inputs and techniques. Example training methods may include, for example, supervised learning, (e.g., decision tree learning, support vector machines, similarity and metric learning, etc.), unsupervised learning, (e.g., association rule learning, clustering, etc.), reinforcement learning, semi-supervised learning, self-supervised learning, multi-instance learning, inductive learning, deductive inference, transductive learning, sparse dictionary learning and the like. Example clustering algorithms used in unsupervised learning may include, for example, k-means clustering, density based special clustering of applications with noise (DBSCAN), mean shift clustering, expectation maximization (EM) clustering using Gaussian mixture models (GMM), agglomerative hierarchical clustering, or the like. According to one embodiment, clustering of data may be performed using a cluster model to group data points based on certain similarities using unlabeled data. Example cluster models may include, for example, connectivity models, centroid models, distribution models, density models, group models, graph based models, neural models and the like.

[0068] One subfield of machine learning includes neural networks, which take inspiration from biological neural networks. In machine learning, a neural network includes interconnected units that process information by responding to external inputs to find connections and derive meaning from undefined data. A neural network can, in a sense, learn to perform tasks by interpreting numerical patterns that take the shape of vectors and by categorizing data based on similarities, without being programmed with any task-specific rules. A neural network generally includes connected units, neurons, or nodes (e.g., connected by synapses) and may allow for the machine learning program to improve performance. A neural network may define a network of functions, which have a graphical relationship. Various neural networks that implement machine learning exist including, for example, feedforward artificial neural networks, perceptron and multilayer perceptron neural networks, radial basis function artificial neural networks, recurrent artificial neural networks, modular neural networks, long short term memory networks, as well as various other neural networks.

[0069] Neural networks may perform a supervised learning process where known inputs and known outputs are utilized to categorize, classify, or predict a quality of a future input. However, additional or alternative embodiments of the machine learning program may be trained utilizing unsupervised or semi-supervised training, where none of the outputs or some of the outputs are unknown, respectively. Typically, a machine learning algorithm is trained (e.g., utilizing a training data set) prior to modeling the problem with which the algorithm is associated. Supervised training of the neural network may include choosing a network topology suitable for the problem being modeled by the network and providing a set of training data representative of the problem.

Generally, the machine learning algorithm may adjust the weight coefficients until any error in the output data generated by the algorithm is less than a predetermined, acceptable level. For instance, the training process may include comparing the generated output produced by the network in response to the training data with a desired or correct output. An associated error amount may then be determined for the generated output data, such as for each output data point generated in the output layer. The associated error amount may be communicated back through the system as an

error signal, where the weight coefficients assigned in the hidden layer are adjusted based on the error signal. For instance, the associated error amount (e.g., a value between -1 and 1) may be used to modify the previous coefficient, e.g., a propagated value. The machine learning algorithm may be considered sufficiently trained when the associated error amount for the output data is less than the predetermined, acceptable level (e.g., each data point within the output layer includes an error amount less than the predetermined, acceptable level). Thus, the parameters determined from the training process can be utilized with new input data to categorize, classify, and/or predict other values based on the new input data.

[0070] The artificial intelligence systems and structures discussed herein may employ deep learning. Deep learning is a particular type of machine learning that provides greater learning performance by representing a certain real-world environment as a hierarchy of increasing complex concepts. Deep learning typically employs a software structure comprising several layers of neural networks that perform nonlinear processing, where each successive layer receives an output from the previous layer. Generally, the layers include an input layer that receives raw data from a sensor, a number of hidden layers that extract abstract features from the data, and an output layer that identifies a certain thing based on the feature extraction from the hidden layers. The neural networks include neurons or nodes that each has a “weight” that is multiplied by the input to the node to obtain a probability of whether something is correct. More specifically, each of the nodes has a weight that is a floating point number that is multiplied with the input to the node to generate an output for that node that is some proportion of the input. The weights are initially “trained” or set by causing the neural networks to analyze a set of known data under supervised processing and through minimizing a cost function to allow the network to obtain the highest probability of a correct output. Deep learning neural networks are often employed to provide image feature extraction and transformation for the visual detection and classification of objects in an image, where a video or stream of images can be analyzed by the network to identify and classify objects and learn through the process to better recognize the objects. Thus, in these types of networks, the system can use the same processing configuration to detect certain objects and classify them differently based on how the algorithm has learned to recognize the objects.

[0071] FIG. 2 illustrates a feedforward neural network **110** that includes a hidden layer **114** between an input layer **112** and an output layer **116**. The input layer **112**, having nodes commonly referenced in FIG. 2 as input nodes **118** for convenience, communicates input data, variables, matrices, or the like to the hidden layer **114**, having nodes **120**. The hidden layer **114** generates a representation and/or transformation of the input data into a form that is suitable for generating output data. Adjacent layers of the neural network **110** are connected at the edges of the nodes of the respective layers, but nodes within a layer typically are not separated by an edge. In at least one embodiment of such a feedforward neural network, data is communicated to the nodes **118** of the input layer **112**, which then communicates the data to the hidden layer **114**. The hidden layer **114** may be configured to determine the state of the nodes in the respective layers and assign weight coefficients or parameters of the nodes based on the edges separating each of the layers, such as an activation function implemented between the input data communicated from the input layer **112** and the output data communicated to nodes **122** of the output layer **116**. It should be appreciated that the form of the output from the neural network may generally depend on the type of model represented by the algorithm. Although the feedforward neural network **110** expressly includes a single hidden layer, other embodiments of feedforward networks within the scope of the descriptions can include any number of hidden layers. The hidden layers are intermediate the input and output layers and are generally where all or most of the computation is done.

[0072] An additional or alternative type of neural network suitable for use in a machine learning program and/or module is a convolutional neural network (CNN). A CNN is a type of feedforward neural network that may be utilized to model data associated with input data having a grid-like topology. In some embodiments, at least one layer of a CNN may include a sparsely connected

layer, in which each output of a first hidden layer does not interact with each input of the next hidden layer. For example, the output of the convolution in the first hidden layer may be an input of the next hidden layer, rather than a respective state of each node of the first layer. CNNs are typically trained for pattern recognition, such as speech processing, language processing, and visual processing. As such, CNNs may be particularly useful for implementing optical and pattern recognition programs required from the machine learning program. A CNN includes an input layer, a hidden layer, and an output layer, typical of feedforward networks, but the nodes of a CNN input layer are generally organized into a set of categories via feature detectors and based on the receptive fields of the sensor, retina, input layer, etc. Each filter may then output data from its respective nodes to corresponding nodes of a subsequent layer of the network. A CNN may be configured to apply the convolution mathematical operation to the respective nodes of each filter and communicate the same to the corresponding node of the next subsequent layer. As an example, the input to the convolution layer may be a multidimensional array of data. The convolution layer, or hidden layer, may be a multidimensional array of parameters determined while training the model.

[0073] FIG. 3 is an illustration of an exemplary CNN **130** that includes an input layer **132** and an output layer **134**. However, where the single hidden layer **114** is provided in the network **110**, multiple consecutive hidden layers **136**, **138** and **140** are provided in the CNN **130**. Edge neurons **142** represented by white-filled arrows highlight that hidden layer nodes **144** can be connected locally, such that not all of the nodes of succeeding layers are connected by neurons.

[0074] FIG. 4 shows a portion of the CNN **130**, specifically portions of the input layer **132** and the first hidden layer **136**, and illustrates that connections can be weighted. In the illustrated example, labels **W1** and **W2** refer to respective assigned weights for the referenced connections. The two hidden nodes **146** and **148** share the same set of weights **W1** and **W2** when connecting to two local patches.

[0075] A weight defines the impact a node in any given layer has on computations by a connected node in the next layer. FIG. 5 shows a network **150** including a node **152** in a hidden layer. The node **152** is connected to several nodes in the previous layer representing inputs to the node **152**. Input nodes **154**, **156**, **158** and **160** in an input layer **162** are each assigned a respective weight **W01**, **W02**, **W03**, and **W04** in the computation at the node **152**, which in this example is a weighted sum.

[0076] An additional or alternative type of feedforward neural network suitable for use in the machine learning program and/or module is a recurrent neural network (RNN). An RNN may allow for analysis of sequences of inputs rather than only considering the current input data set. RNNs typically include feedback loops/connections between layers of the topography, thus allowing parameter data to be communicated between different parts of the neural network. RNNs typically have an architecture including cycles, where past values of a parameter influence the current calculation of the parameter, e.g., at least a portion of the output data from the RNN may be used as feedback/input in calculating subsequent output data. In some embodiments, the machine learning module may include an RNN configured for language processing, e.g., an RNN configured to perform statistical language modeling to predict the next word in a string based on the previous words. The RNN(s) of the machine learning program may include a feedback system suitable to provide the connection(s) between subsequent and previous layers of the network.

[0077] FIG. 6 illustrates an RNN **170** that includes an input layer **172** with nodes **174**, an output layer **176** with nodes **178**, and multiple consecutive hidden layers **180** and **184** with nodes **182** and nodes **186**, respectively. The RNN **170** also includes a feedback connector **188** configured to communicate parameter data from at least one of the nodes **186** in the second hidden layer **184** to at least one of the nodes **182** in the first hidden layer **180**. It should be appreciated that two or more and up to all of the nodes of a subsequent layer may provide or communicate a parameter or other data to a previous layer of the RNN **170**. Moreover and in some embodiments, the RNN **170** may

include multiple feedback connectors, such as connectors suitable to communicatively couple pairs of nodes and/or connector systems configured to provide communication between three or more nodes. Additionally or alternatively, the feedback connector **188** may communicatively couple two or more nodes having at least one hidden layer between them, i.e., nodes of non-sequential layers of the RNN **170**.

[0078] In an additional or alternative embodiment, the machine learning program may include one or more support vector machines. A support vector machine may be configured to determine a category to which input data belongs. For example, the machine learning program may be configured to define a margin using a combination of two or more of the input variables and/or data points as support vectors to maximize the determined margin. Such a margin may generally correspond to a distance between the closest vectors that are classified differently. The machine learning program may be configured to utilize a plurality of support vector machines to perform a single classification. For example, the machine learning program may determine the category to which input data belongs using a first support vector determined from first and second data points/variables, and the machine learning program may independently categorize the input data using a second support vector determined from third and fourth data points/variables. The support vector machine(s) may be trained similarly to the training of neural networks, e.g., by providing a known input vector (including values for the input variables) and a known output classification. The support vector machine is trained by selecting the support vectors and/or a portion of the input vectors that maximize the determined margin.

[0079] As depicted, and in some embodiments, the machine learning program may include a neural network topography having more than one hidden layer. In such embodiments, one or more of the hidden layers may have a different number of nodes and/or the connections defined between layers. In some embodiments, each hidden layer may be configured to perform a different function. As an example, a first layer of the neural network may be configured to reduce a dimensionality of the input data, and a second layer of the neural network may be configured to perform statistical programs on the data communicated from the first layer. In various embodiments, each node of the previous layer of the network may be connected to an associated node of the subsequent layer (dense layers). Generally, the neural network(s) of the machine learning program may include a relatively large number of layers, e.g., three or more layers, and are referred to as deep neural networks. For example, the node of each hidden layer of a neural network may be associated with an activation function utilized by the machine learning program to generate an output received by a corresponding node in the subsequent layer. The last hidden layer of the neural network communicates a data set (e.g., the result of data processed within the respective layer) to the output layer. Deep neural networks may require more computational time and power to train, but the additional hidden layers provide multistep pattern recognition capability and/or reduced output error relative to simple or shallow machine learning architectures (e.g., including only one or two hidden layers).

[0080] According to various implementations, deep neural networks incorporate neurons, synapses, weights, biases, and functions and can be trained to model complex non-linear relationships. Various deep learning frameworks may include, for example, TensorFlow, MxNet, PyTorch, Keras, Glue, and the like. Training a deep neural network may include complex input/output transformations and may include, according to various embodiments, a backpropagation algorithm. According to various embodiments, deep neural networks may be configured to classify images of handwritten digits from a dataset or various other images. According to various embodiments, the datasets may include a collection of files that are unstructured and lack predefined data model schema or organization. Unlike structured data, which is usually stored in a relational database (RDBMS) and can be mapped into designated fields, unstructured data comes in many formats that can be challenging to process and analyze. Examples of unstructured data may include, according to non-limiting examples, dates, numbers, facts, emails, text files, scientific data, satellite imagery,

media files, social media data, text messages, mobile communication data, and the like.

[0081] FIG. 7 is a block diagram of an artificial intelligence programming system **200** including an AI processor **202**, such as a dedicated processing device, that operates an artificial intelligence program, where the processor **202** includes a front-end sub-processor **204** and a back-end sub-processor **206**. The algorithms associated with the front-end sub-processor **204** and the back-end sub-processor **206** may be stored in an associated memory device and/or storage device, such as memory device **208** communicatively coupled to the AI processor **202**, as shown. Additionally, the system **200** may include a memory **212** storing one or more instructions necessary for operating the AI program. In this embodiment, the sub-processor **204** includes neural networks **214** and **216** operating an AI algorithm **218**, such as feature recognition, and the sub-processor **206** includes neural networks **220** and **222** operating an AI algorithm **224** to perform an operation on the data set communicated directly or indirectly to the sub-processor **206**.

[0082] The system **200** may provide statistical models or machine learning programs such as decision tree learning, associate rule learning, recurrent artificial neural networks, support vector machines, and the like. In various embodiments, the sub-processor **204** may be configured to include built in training and inference logic or suitable software to train the neural network prior to use, for example, machine learning logic including, but not limited to, image recognition, mapping and localization, autonomous navigation, speech synthesis, document imaging, or language translation, such as natural language processing. For example, the sub-processor **204** may be used for image recognition, input categorization, and/or support vector training. In various embodiments, the sub-processor **206** may be configured to implement input and/or model classification, speech recognition, translation, and the like.

[0083] For instance and in some embodiments, the system **200** may be configured to perform unsupervised learning, in which the machine learning program performs the training process using unlabeled data, e.g., without known output data with which to compare. During such unsupervised learning, the neural network may be configured to generate groupings of the input data and/or determine how individual input data points are related to the complete input data set. For example, unsupervised training may be used to configure a neural network to generate a self-organizing map, reduce the dimensionality of the input data set, and/or to perform outlier/anomaly determinations to identify data points in the data set that falls outside the normal pattern of the data. In some embodiments, the system **200** may be trained using a semi-supervised learning process in which some but not all of the output data is known, e.g., a mix of labeled and unlabeled data having the same distribution.

[0084] In some embodiments, the system **200** may include an index of basic operations, subroutines, and the like (primitives) typically implemented by AI and/or machine learning algorithms. Thus, the system **200** may be configured to utilize the primitives of the processor **202** to perform some or all of the calculations required by the system **200**. Primitives suitable for inclusion in the processor **202** include operations associated with training a convolutional neural network (e.g., pools), tensor convolutions, activation functions, basic algebraic subroutines and programs (e.g., matrix operations, vector operations), numerical method subroutines and programs, and the like.

[0085] It should be appreciated that the machine learning program may include variations, adaptations, and alternatives suitable to perform the operations necessary for the system, and the present disclosure is equally applicable to such suitably configured machine learning and/or artificial intelligence programs, modules, etc. For instance, the machine learning program may include one or more long short-term memory (LSTM) RNNs, convolutional deep belief networks, deep belief networks DBNs, and the like. DBNs, for instance, may be utilized to pre-train the weighted characteristics and/or parameters using an unsupervised learning process. Further, the machine learning module may include one or more other machine learning tools (e.g., logistic regression (LR), Naive-Bayes, random forest (RF), matrix factorization, and support vector

machines) in addition to, or as an alternative to, one or more neural networks, as described herein. [0086] FIG. **8** is a flow chart diagram **230** showing an exemplary method for model development and deployment by machine learning. The method represents at least one example of a machine learning workflow in which steps are implemented in a machine learning project. At box **232**, a user authorizes, requests, manages, or initiates the machine-learning workflow. This may represent a user such as human agent, or customer, requesting machine-learning assistance or AI functionality to simulate intelligent behavior, such as a virtual agent, or other machine-assisted or computerized tasks that may, for example, entail visual perception, speech recognition, decision-making, translation, forecasting, predictive modelling, and/or suggestions as non-limiting examples. In a first iteration from the user perspective, the box **232** can represent a starting point. However, with regard to continuing or improving an ongoing machine learning workflow, the box **232** can represent an opportunity for further user input or oversight via a feedback loop.

[0087] At box **234**, data is received, collected, accessed or otherwise acquired and entered as can be termed data ingestion. At box **236**, data ingested from the box **234** is pre-processed, for example, by cleaning, and/or transformation such as into a format that the following components can digest. The incoming data may be versioned to connect a data snapshot with the particularly resulting trained model. As newly trained models are tied to a set of versioned data, preprocessing steps are tied to the developed model. If new data is subsequently collected and entered, a new model will be generated. If the preprocessing is updated with newly ingested data, an updated model will be generated. The process at the box **236** can include data validation, which focuses on confirming that the statistics of the ingested data are as expected, such as that data values are within expected numerical ranges, that data sets are within any expected or required categories, and that data comply with any needed distributions such as within those categories. The process can proceed to box **238** to automatically alert the initiating user, other human or virtual agents, and/or other systems, if any anomalies are detected in the data, thereby pausing or terminating the process flow until corrective action is taken.

[0088] At box **240**, training test data, such as a target variable value, is inserted into an iterative training and testing loop. At box **242**, model training, a core step of the machine learning workflow, is implemented. A model architecture is trained in the iterative training and testing loop. For example, features in the training test data are used to train the model based on weights and iterative calculations in which the target variable may be incorrectly predicted in an early iteration as determined by comparison at box **244**, where the model is tested. Subsequent iterations of the model training at the box **242** may be conducted with updated weights in the calculations.

[0089] When compliance and/or success in the model testing at the box **244** is achieved, the process proceeds to box **246**, where model deployment is triggered. The model may be utilized in AI functions and programming, for example, to simulate intelligent behavior, to perform machine-assisted or computerized tasks, of which visual perception, speech recognition, decision-making, translation, forecasting, predictive modelling, and/or automated suggestion generation serve as non-limiting examples.

[0090] FIG. **9** is a simple illustration of a typical home or landing page **250** for an online or mobile banking application that opens on any suitable user device as described above when a bank customer or user logs into the application. From the landing page **250**, the user can select or click-on a number of buttons or tabs to get to the desired area in the application to perform certain tasks, such as observing balances, paying bills, transferring money, etc. In this non-limiting example, the landing page **250** includes an Accounts button **252**, a Bill Pay button **254**, a Transfer Funds button **256**, a Customer Service button **260**, a user Profile button **262**, a Contact button **264** and a Save Money button **266**, where clicking each button will direct the user to various application pages from which selections can be made to perform various tasks related to that selection. This example of the landing page **250** shows a summary of account balances.

[0091] Clicking the Accounts button **252** will, for example, direct the user to application pages that

allow the user to view his/her various accounts. Clicking the Bill pay button **254** will, for example, direct the user to application pages that allow the user to pay previously loaded vendors whose bills can be paid either electronically or by mail. Clicking the Transfer Funds button **256** will, for example, direct the user to an application pages that allow the user to transfer funds between internal accounts for that institution, to transfer funds to an account outside of that institution that has been previously verified, wire funds to an international account and transfer funds to a person using a third party application, such as Zelle™. Clicking the Customer Service button **260** will, for example, direct the user to application pages that allow the user to take advantage of various banking services. Clicking the Profile button **262** can, for example, direct the user to application pages that allow the user to view and change his/her settings, such as phone number, address, email address, beneficiaries, etc. Clicking the Contact button **264** will, for example, direct the user to application pages that allow the user to view the institutions contact and location information. [0092] Selecting the Save Money button **266** may open a page, for example, page **270** shown in FIG. **10** that will provide recommendations to the user as to where to reduce spending in response to the user identifying how much he/she wants to save and considering user preferences for not reducing spending, i.e., the areas that are important to the user for maintaining spending. For example, the user may be overly fond of coffee and not want to reduce their spending at Starbucks™. The back-end of the application can obtain data and information from many sources to identify where, for what, how often and how much the user is currently spending on various things or categories of things, such as dining out, travel, entertainment, groceries, etc. The application will look at the things the user is spending money on including when, where and how much the user is spending, and the things that the user has identified as being spending areas that are important to the user, and will provide recommendations for reducing spending first in those areas that may be less important to the user and ultimately to areas that are more important to the user. In other words, based on how much the user wants to save and where the user would not prefer to reduce spending, the application will provide recommendations for reducing spending in a selective manner first considering the less important areas and then considering the more important areas. The application may allow the user to prioritize the areas not to reduce spending. The application can be programmed to provide more or less user input into the process for saving money as desired.

[0093] The page **270** may provide a button **272** that when clicked asks the user how much the user wants to save, which may be selected from a drop down list **274** of amounts. Further, the page **270** may provide a button **276** that when clicked asks the user to select things that are important to the user, where the user may not want to reduce his/her spending, and also may be in the form of a drop down list **278** including things such as travel, fitness, dining, movies, groceries, beauty products, coffee, etc. The user can select those things that are important to the user from the list **278**, and can prioritize those things that are important to the user. Alternately, the application can ask the user how much he/she wants to save and what areas would they prefer not to reduce their spending. The application can provide recommendations to the user in a window **280** of where the user can reduce spending based on the user's current spending habits and the things that are less important to the user. The things that appear in the window **280** can be categorized in an order that highlights the areas of spending that may be less important to the user. Further, the back-end of the application can monitor the user's spending habits through credit card and debit card transactions and periodically provide a pop up window **282** on the user's device, such as a mobile phone, of where the user could have reduced his/her spending.

[0094] In an alternate embodiment, the user may engage or open a bot, i.e., an algorithm that the user can correspond with and obtain information from, to provide the recommendations for reducing spending as discussed above. FIG. **11** is a simple illustration of a page **290** showing the user conversing with the bot in a window **292** for this purpose.

[0095] The online or mobile banking application described above may employ some of the

processors and neural networks also described above to perform the various processes for providing recommendations to a user for saving money and/or reducing spending, as described. FIG. 12 is a block diagram of an architecture 300 that could be part of the enterprise system 12 that could perform these operations. The architecture 300 includes a repository 302 having a plurality of databases 304 that store data and information in a format accessible to users, and a back-end server 306 operatively coupled to the repository 302 and being responsive to the data and information from all of the databases 304. The back-end server 306 includes a processor 308 for processing the data and information, a communications interface 310 communicatively coupled to the processor 308 and a memory device 312 for storing data and executable code. The executable code causes the processor 308 to collect data and information from the databases 304, store the collected data and information in the memory device 312, process the stored data and information through a machine learning model, receive a result from the machine learning model, and transmit a communication identifying the result on the interface 310.

[0096] Particular embodiments and features have been described with reference to the drawings. It is to be understood that these descriptions are not limited to any single embodiment or any particular set of features. Similar embodiments and features may arise or modifications and additions may be made without departing from the scope of these descriptions and the spirit of the appended claims.

Claims

1. A method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save, said method comprising: allowing the user to enter how much the user wants to save; allowing the user to enter information concerning areas that the user does not want to reduce spending; determining what the user is spending on, where the user is spending, how much the user is spending and how often the user is spending; providing recommendations to the user of areas that the user can reduce spending considering the amount the user wants to save and the areas that the user does not want to reduce spending; and prioritizing the recommendations for reducing spending by first recommending areas for reducing spending on things that the user has not identified as being areas that the user does not want to reduce spending and then recommending areas for reducing spending on areas that the user does not want to reduce spending, wherein prioritizing the recommendations for reducing spending includes using machine learning and at least one neural network, and wherein the at least one neural network includes nodes that have been trained to prioritizes the recommendations for reducing spending, and wherein the nodes in the at least one neural network have been trained by training nodes in a neural network simulation model that employs unsupervised learning and training data to prioritizes the recommendations for reducing spending and a target variable, wherein the unsupervised learning is used to configure the neural network to generate a self-organizing map, reduce the dimensionality of an input data set, and to perform outlier/anomaly determinations to identify data points in the data set that falls outside a normal pattern of the data, and wherein training the nodes in the simulation model employs an iterative training and testing loop that incorporates weights associated with the nodes in the simulation model and iterative calculations that are tested, compared to the target variable and updated in subsequent iterative calculations to improve predictability of the target variable.
2. The method according to claim 1 wherein determining what the user is spending on is based on user credit card spending, debit card spending and/or bill paying.
3. The method according to claim 1 wherein determining where the user is spending, how much the user is spending and how often the user is spending on dining out, travel, fitness, beauty products, entertainment and groceries.
4. The method according to claim 1 further comprising allowing the user to prioritize the areas that

the user does not want to reduce spending.

5. The method according to claim 1 further comprising providing a pop-up window for where the user can reduce spending in response to credit card and debit card purchases.

6. A method for providing recommendations to a user for reducing spending in response to the user identifying how much he/she wants to save, said method comprising: allowing the user to enter how much the user wants to save; allowing the user to enter information concerning areas that the user does not want to reduce spending; determining what the user is spending on, where the user is spending, how much the user is spending and how often the user is spending; providing recommendations to the user of areas that the user can reduce spending considering the amount the user wants to save and the areas that the user does not want to reduce spending; and prioritizing the recommendations for reducing spending by first recommending areas for reducing spending on things that the user has not identified as being areas that the user does not want to reduce spending and then recommending areas for reducing spending on areas that the user does not want to reduce spending.

7. The method according to claim 6 wherein determining what the user is spending on is based on user credit card spending, debit card spending and/or bill paying.

8. The method according to claim 6 wherein determining where the user is spending, how much the user is spending and how often the user is spending on dining out, travel, fitness, beauty products, entertainment and groceries.

9. The method according to claim 6 further comprising allowing the user to prioritize the areas that the user does not want to reduce spending.

10. The method according to claim 6 further comprising providing a pop-up window for where the user can reduce spending in response to credit card and debit card purchases.
